

BSc (Hons) Artificial Intelligence and Data Science

Module: CM1601 Programming Fundamentals

Individual Coursework Report

RGU Student ID : 2506755

IIT Student ID : 20240281

Student Name : W.D. Nisitha Nimsara

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3.Flowchart – Function 1 (AID- Adding item details)

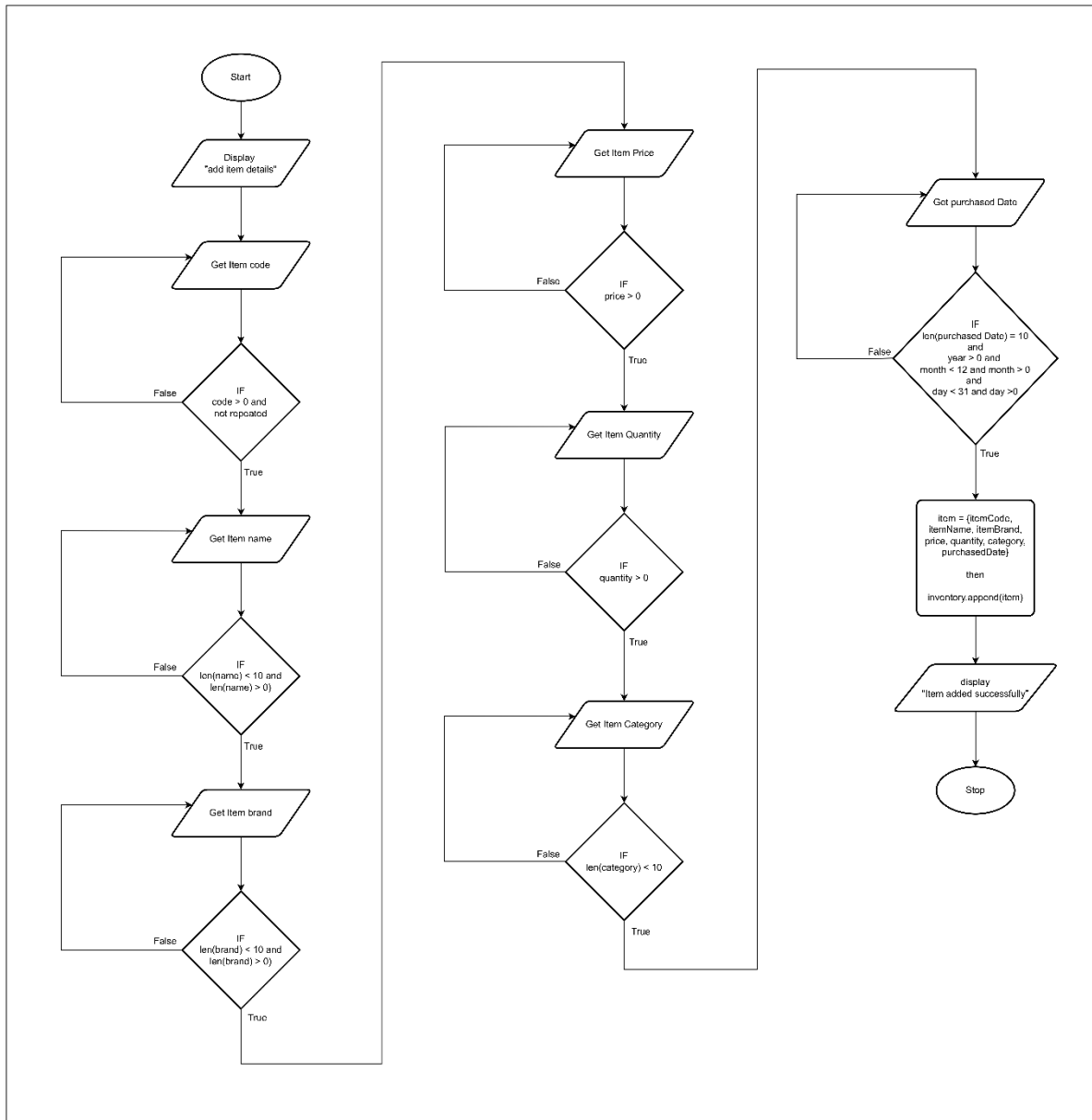


Figure 1 - Flowchart of AID function

4.Pseudocode – Function 1 (AID- Adding item Details)

Start

Display "Add item details"

Repeat

 Prompt "Enter item code:"

 Get itemCode

Until itemCode > 0 AND itemCode is not repeated

Repeat

 Prompt "Enter item name:"

 Get itemName

Until length(itemName) < 10 AND length(itemName) > 0

Repeat

 Prompt "Enter item brand:"

 Get itemBrand

Until length(itemBrand) < 10 AND length(itemBrand) > 0

Repeat

 Prompt "Enter item price:"

 Get price

Until price > 0

Repeat

 Prompt "Enter item quantity:"

 Get quantity

Until quantity > 0

Repeat

 Prompt "Enter item category:"

 Get category

Until length(category) < 10

Repeat

 Prompt "Enter purchased date (YYYY-MM-DD):"

 Get purchasedDate

 Split purchasedDate into year, month, day

Until length(purchasedDate) == 10 AND

 year > 0 AND

 1 <= month <= 12 AND

 1 <= day <= 31

Create

item = {itemCode, itemName, itemBrand, price, quantity, category, purchasedDate}

Add item to inventory

Display "Item added successfully"

Stop

5.Flowchart – Function 2 (SDD- selecting four dealers randomly from a file)

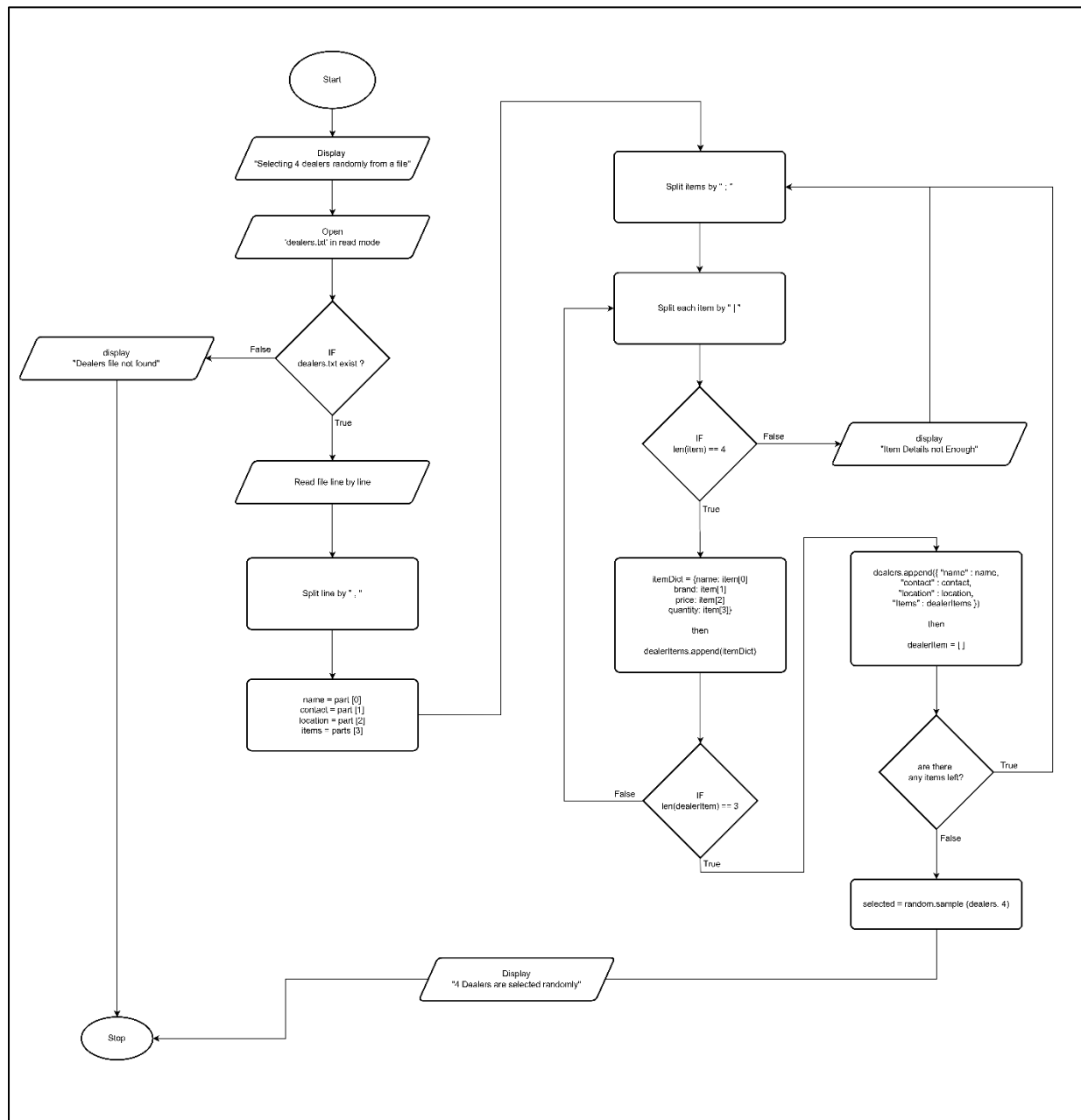


Figure 2 - Flowchart of SDD function

6.Pseudocode – Function 2 (SDD- selecting four dealers randomly from a file)

Start

Display "Selecting 4 dealers randomly from a file"
Open "dealers.txt" in read mode

If file exists :

For line in file:

Split line by " , "
name = part[0]
contact = part[1]
location = part[2]
items = part[3]

Split items by " ; "

For each item in items:

Split item by " | "

If length of item == 4 :

itemDict = {
 "name": item[0],
 "brand": item[1],
 "price": float(item[2]),
 "quantity": int(item[3])
}

Append itemDict to dealerItems

If length of dealerItems == 3 :

Append {
 "name": name,
 "contact": contact,
 "location": location,
 "items": dealerItems
} to dealers
Clear dealerItems

Else:

Display "Item Details not Enough"

selected = random.sample(dealers, 4)

Display "4 Dealers are selected randomly"

Return selected

Else:

Display "Dealers file not found"

Stop

7.Source Code

```
import random
#Create variables
choice = 0
state = True
try_again = 0
itemCode = 0
inventory = []
invenCheck = []
dealers=0
fileSave = None

def sorting():
    # sorting inventory according to the Item Code value
    length = len(inventory)

    for key in range(length):
        lowValue = key
        for newKey in range(key + 1, length):
            if int(inventory[newKey]['Item Code']) < int(inventory[lowValue]['Item Code']):
                lowValue = newKey
        # Change elements
        inventory[key], inventory[lowValue] = inventory[lowValue], inventory[key]

def date(givedate):
    "This function help to get date correctly "
    year = givedate[0: 4]
    month = givedate[5: 7]
    day = givedate[8:10]
    realdate = year + "/" + month + "/" + day
    return realdate

def displayMenu():
    "This function will Display the possible command manu."
    print("""
    Type AID for adding item details.
    Type DID for deleting item details.
    Type UID for updating item details.
    Type VID for viewing the items table.
    Type SID for saving the item details to the text file.
    Type SDD for selecting four dealers randomly from a file.
    Type VRL for displaying all the details of the randomly selected dealers.
    Type LDI for displaying the items of the given dealer.
    Type ESC to exit the program.
    """)

def addItemDetails():
    "This function get item details and store in inventory."
```

```

itemName = None
itemBrand = None
price = 0.0
quantity = 0
category = None
purchasedDate = None
item = None
print("Adding item details")
while True: # For get correct Item code
    try:
        #getting input from user
        itemCode = int(input("Enter Item Code (3 Digit): "))
        itemCode = f"{itemCode:03d}" #converting 3 digit
        if int(itemCode) < 0:
            print("Invalid Item Code! Try again..")
        elif invenCheck.count(itemCode) == 1:
            print("This item code already Registered!")
        else:
            invenCheck.append(itemCode)
            break
    except ValueError or TypeError:
        print("Invalid Format!!!")

while True: # get correct item name
    itemName = input("Enter Item Name: ")
    if len(itemName) > 10 or len(itemName) <= 0:
        print("Invalid item name! Try again..")
    else:
        break

while True: # to get correct item brand
    itemBrand = input("Enter Item Brand: ")
    if len(itemBrand) > 10 or len(itemBrand) <= 0:
        print("Invalid item Brand! Try again..")
    else:
        break

while True: # For get correct Item Price
    try:
        price = float(input("Enter Item Price: "))
        if price <= 0:
            print("Invalid Price! Try again..")
        else:
            break
    except ValueError or TypeError:
        print("Invalid Format!!!")

while True: # For get correct Item Quantity
    try:
        quantity = int(input("Enter Quantity: "))

```

```

        if quantity < 0:
            print("Invalid Quantity! Try again..")
        else:
            break
    except ValueError or TypeError:
        print("Invalid Format!!!")

while True:                # to get correct item Category
    category = input("Enter Item Category: ")
    if len(category) > 10:
        print("Invalid item Category! try again..")
    else:
        break

while True: # to get correct item Purchased Date
    try:
        Date = str(input("Enter Purchased Date (YYYY/MM/DD): "))
        year = int(Date[0:4])
        month = int(Date[5:7])
        day = int(Date[8:10])
        if len(Date) != 10 or year <= 0 or month > 12 or month <= 0 or day > 31 or day <= 0:
            print("Invalid Purchased Date! try again..")
        else:
            purchasedDate = date(Date)
            break
    except ValueError:
        print("Invalid Format!!!")

item = {
    'Item Code': itemCode,
    'Item Name': itemName,
    'Item Brand': itemBrand,
    'Price': price,
    'Quantity': quantity,
    'Category': category,
    'Purchased Date': purchasedDate}
inventory.append(item)
print("Item added successfully!")

def deleteitemDetails():
    "This function delete item from inventory"
    print("Deleting item details")
    while True:
        itemCode = int(input("Enter Item Code to delete: "))
        itemCode = f"{itemCode:03d}" # converting 3 digit
        for item in inventory:
            if item['Item Code'] == itemCode:
                inventory.remove(item)
                print("Item deleted successfully!")

```

```

        return
    else:
        print("Item not found!")

def updateItemDetails():
    "This function for update item detail other than item code."
    print("Updating item details")
    try:
        while True:
            itemCode = int(input("Enter Item Code to update: "))
            itemCode = f'{itemCode:03d}' # converting 3 digit
            for item in inventory:
                if item['Item Code'] == itemCode:
                    while True: # update item name
                        newItemName = input("Enter Item Name: ")
                        if len(newItemName) > 15 or len(newItemName) <= 0:
                            print("Invalid item name! Try again..")
                        else:
                            item['Item Name'] = newItemName
                            break

                    while True: # update item brand
                        newItemBrand = input("Enter Item Brand: ")
                        if len(newItemBrand) > 15 or len(newItemBrand) <= 0:
                            print("Invalid item Brand! Try again..")
                        else:
                            item['Item Brand'] = newItemBrand
                            break

                    while True: # update Item Price
                        try:
                            newPrice = float(input("Enter Item Price: "))
                            if newPrice <= 0:
                                print("Invalid Price! Try again..")
                            else:
                                item['Price'] = newPrice
                                break
                        except ValueError or TypeError:
                            print("Invalid Format!!!")

                    while True: # update Item Quantity
                        try:
                            newQuantity = int(input("Enter Quantity: "))
                            if newQuantity < 0:
                                print("Invalid Quantity! Try again..")
                            else:
                                item['Quantity'] = newQuantity
                                break
                        except ValueError or TypeError:
                            print("Invalid Format!!!")

```

```

while True: # update item Category
    newCategory = input("Enter Item Category: ")
    if len(newCategory) > 15:
        print("Invalid item Category! try again..")
    else:
        item['Category'] = newCategory
        break

while True: # update item Purchased Date
    try:
        newDate = str(input("Enter Purchased Date (YYYY/MM/DD): "))
        year = int(newDate[0:4])
        month = int(newDate[5:7])
        day = int(newDate[8:10])
        if len(newDate) != 10 or year <= 0 or month > 12 or month <= 0 or day >
31 or day <= 0:
            print("Invalid Purchased Date! try again..")
        else:
            newPurchasedDate = date(newDate)
            item['Purchased Date'] = newPurchasedDate
            break
    except ValueError:
        print("Invalid Format!!!")
        print("Item updated successfully!")
        return
    else:
        print("Item not found!")
except ValueError:
    print("Try Again...")

def viewItemDetails():
    'This function will display Item Details as a table and Total value od items'
    print("Viewing the items table and current total")
    #display tables Headings
    print("| ", "Item Code", " " * (10 - len("Item Code")), " | ", "Item Name", " " * (10 -
len("Item Name")), " | ", "Item Brand",
        " " * (10 - len("Item Brand")), " | ", "Price(Rs.)", " " * (10 - len("Price(Rs.)")), " | ",
"Quantity",
        " " * (10 - len(str("Quantity"))), " | ", "Category", " " * (10 - len("Category")), " | ",
"Purchased Date",
        " " * (15 - len("Purchased Date")), " | ")
    #Display table
    for x in inventory:
        print("| ", x["Item Code"], " " * (10 - len(x["Item Code"])), " | ", x["Item Name"], " " *
(10 - len(x["Item Name"])), " | ", x["Item Brand"],
            " " * (10 - len(x["Item Brand"])), " | ", x["Price"], " " * (10 - len(str(x["Price"]))), " |
", x["Quantity"],
            " " * (10 - len(str(x["Quantity"]))), " | ", x["Category"], " " * (10 -
len(x["Category"])), " | ",

```

```

        x["Purchased Date"], " " * (15 - len(x["Purchased Date"])), " |")
    total = sum(item['Price'] * item['Quantity'] for item in inventory)
    print(f'Total value of items: Rs {total:.2f}')

```

```
def saveItemDetails():
```

```
    'This function will save item details to a text file.'
```

```
    # This will open the file in Writing mode
```

```
    with open('Item Details.txt','w') as file:
```

```
        heading = ["Item Code", "Item Name", "Item Brand", "Price", "Quantity", "Category",
"Purchased Date"]
```

```
        for i in heading:
```

```
            y = str(" " * (10 - len(i)))
```

```
            file.write(f'| {i} {y}')
```

```
        file.write("\n")
```

```
    # This will save item Details in the file
```

```
    with open("Item Details.txt",'a') as file:
```

```
        for dic in inventory:
```

```
            for item in dic:
```

```
                key = item
```

```
                value = dic[key]
```

```
                r = str(" " * (10 - len(str(value))))
```

```
                file.write(f'| {value} {r}')
```

```
            file.write(" |\n")
```

```
    print("\nItem Details Saved to a File!")
```

```
def selectRandomDealers():
```

```
    'This function will select 4 dealers form text file'
```

```
    dealers = []
```

```
    dealerItems = []
```

```
    print("Selecting four dealers randomly from a file")
```

```
    try:
```

```
        with open("dealers.txt","r") as file2:    #Opening text file in read mode
```

```
            for line in file2:
```

```
                parts = line.strip().split(",")    #Spliting using "," mark
```

```
                name, contact, location, items = parts[0], parts[1], parts[2], parts[3]
```

```
            #Dealer's Item details separating
```

```
            for item in items.split(";"):    #Spliting using ";" mark
```

```
                itemParts = item.strip().split("|")    #Spliting using "|" mark
```

```
            if len(itemParts) == 4: #Seperating item parts
```

```
                itemDict = {
```

```
                    "name": itemParts[0],
```

```
                    "brand": itemParts[1],
```

```
                    "price": float(itemParts[2]),
```

```
                    "quantity": int(itemParts[3])}
```

```
                dealerItems.append(itemDict) #append item parts to item Dict
```

```

        if len(dealerItems) == 3: #appending Dealer details to a list
            dealers.append({
                "name": name,
                "contact": contact,
                "location": location,
                "items": dealerItems})
            dealerItems = []

    else:
        print("Item details not Enough!")

    #Selecting 4 random dealers from all dealers
    selected = random.sample(dealers, 4)
    print("4 Dealers are Selected Randomly.")
    return selected
except FileNotFoundError:
    print("Dealers file not found")

def viewRandomDealers(dealers):
    "This function will display the dealer's details"
    print("Displaying all the details of the randomly selected dealers")
    if dealers == 0:
        print("First you need to select Dealers.")
    else:
        print("--Dealer's Details--")
        # display tables Headings
        print("| ", "Name", " " * (10 - len("Name")), " | ", "Contact No", " " * (12 - len("Contact
No")), " | ",
              "Location", " " * (10 - len("Location")), " | ")

        # Sorting according to the location name (A to Z)
        n = len(dealers)
        for i in range(n):
            for j in range(0, n - i - 1):
                if dealers[j]["location"].lower() > dealers[j + 1]["location"].lower():
                    # Swap dealers
                    dealers[j], dealers[j + 1] = dealers[j + 1], dealers[j]

        # Display table
        for x in dealers:
            print("| ", x["name"], " " * (10 - len(x["name"])), " | ", x["contact"],
                  " " * (12 - len(x["contact"])), " | ", x["location"],
                  " " * (10 - len(x["location"])), " | ")

def displayItemsOfDealer(dealers):
    "This function will display the dealer's item details"
    available = None
    print("Display the items of the given dealer")
    try:

```

```

chooseDealer = input("Enter Dealer's Name : ") #Getting input from user

for x in dealers:
    if chooseDealer == x['name'] :          #Selecting the relevant item Details
        available = 1
        print("--Dealer's Item Details--")
        print("|", "Name", " " * (10 - len("Name")), "|", "Brand", " " * (10 - len("Brand")),
"|",
            "Price(Rs.)", " " * (10 - len("Price(Rs.)")), "|", "Quantity", " " * (10 -
len("Quantity")), "|")
        for ss in x['items']:
            for ee in ss:
                print('|',ss[ee]," " * (10 - len(str(ss[ee]))),end=' ')
            print('|','r')
        if available != 1:
            print('Dealer Not Found!')
except TypeError:
    print("Wrong Format Try Again!")

```

#-----Main Program-----

```

while state:
    displayMenu()
    choice = input("Enter your choice: ").upper()
    #Selection option
    if choice == "AID":
        addItemDetails()
        sorting()
        continue
    elif choice == "DID":
        deleteitemDetails()
        continue
    elif choice == "UID":
        updateItemDetails()
        continue
    elif choice == "VID":
        viewItemDetails()
        continue
    elif choice == "SID":
        saveItemDetails()
        continue
    elif choice == "SDD":
        dealers = selectRandomDealers()
        continue
    elif choice == "VRL":
        viewRandomDealers(dealers)
        continue
    elif choice == "LDI":
        displayItemsOfDealer(dealers)

```

```
        continue
    elif choice == "ESC":
        print("Have a nice day !!!")
        break
    else:
        print("Invalid choice.")

    #Try again option.
    try_again = str(input("Do you want to try again: ")).upper()
    if try_again == "NO":
        print("Have a nice day !!!")
        break
    elif try_again == "YES":
        continue
    else:
        print("Assume you want to try again.")
        continue
```

8.Outputs

8.1AID Function

Input

```
"C:\Users\Nisitha Nimsara\AppData\Local\Programs\Python\Python312\python.exe" "C:\Us

Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.

Enter your choice: aid
Adding item details
Enter Item Code (3 Digit): 002
Enter Item Name: laptop
Enter Item Brand: hp
Enter Item Price: 150000
Enter Quantity: 3
Enter Item Category: laptop
Enter Purchased Date (YYYY/MM/DD): 2025/02/11|
```

Figure 3 - AID Function Input

Output

```
Enter Quantity: 3
Enter Item Category: laptop
Enter Purchased Date (YYYY/MM/DD): 2025/02/11
Item added successfully!
```

Figure 4 - AID Function Output

8.2DID Function

Input

```
Type AID for adding item details.  
Type DID for deleting item details.  
Type UID for updating item details.  
Type VID for viewing the items table.  
Type SID for saving the item details to the text file.  
Type SDD for selecting four dealers randomly from a file.  
Type VRL for displaying all the details of the randomly selected dealers.  
Type LDI for displaying the items of the given dealer.  
Type ESC to exit the program.  
  
Enter your choice: did  
Deleting item details  
Enter Item Code to delete: 002
```

Figure 5 - DID Function Input

Output

```
Enter Item Code to delete: 002  
Item deleted successfully!
```

Figure 6 - DID Function Output

8.3 UID Function

Previous details

```
Enter your choice: aid
Adding item details
Enter Item Code (3 Digit): 001
Enter Item Name: zenbook
Enter Item Brand: asus
Enter Item Price: 123456
Enter Quantity: 000
Enter Item Category: laptop
Enter Purchased Date (YYYY/MM/DD): 2025/03/20
```

Figure 7 - Before UID Function

Input

```
Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.

Enter your choice: uid
Updating item details
Enter Item Code to update: 001
Enter Item Name: ZENBOOK
Enter Item Brand: ASUS
Enter Item Price: 250000
Enter Quantity: 4
Enter Item Category: laptop
Enter Purchased Date (YYYY/MM/DD): 2025/03/20
```

Figure 8 - UID Function Input

Output

```
Enter Item Category: laptop
Enter Purchased Date (YYYY/MM/DD): 2025/03/20
Item updated successfully!
```

Figure 9 - UID Function Output

8.4VID Function

Input

```
Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.
```

```
Enter your choice: vid
```

Figure 10 - VID Function Input

Output

```
Enter your choice: vid
Viewing the items table and current total
| Item Code | Item Name | Item Brand | Price(Rs.) | Quantity | Category | Purchased Date |
| 001      | ZENBOOK  | ASUS      | 250000.0   | 4        | laptop   | 2025/03/20     |
| 002      | air4pro   | soundpeats | 13000.0    | 6        | erabuds  | 2024/04/30     |
| 022      | iphone11  | apple     | 180000.0   | 2        | phone    | 2020/09/12     |
Total value of items: Rs 1438000.00
```

Figure 11 - VID Function Output

8.5SID Function

Input

```
Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.

Enter your choice: sid
```

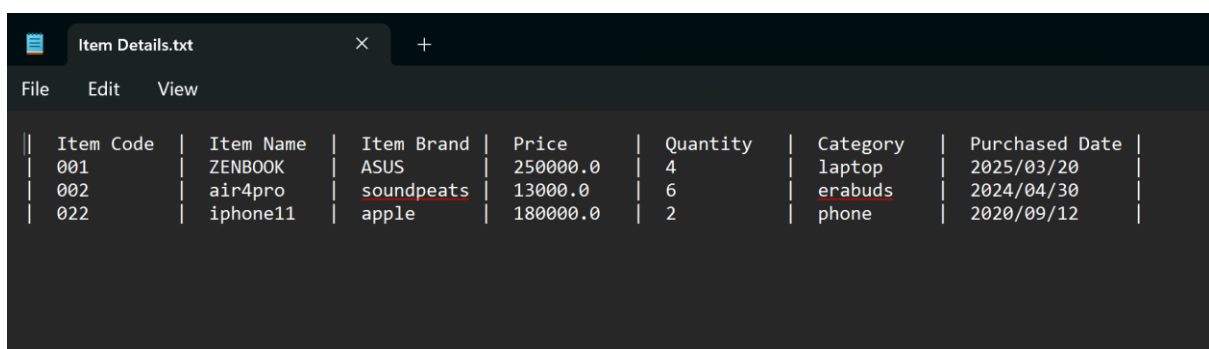
Figure 12 - SID Function Input

Output

```
Enter your choice: sid

Item Details Saved to a File!
```

Figure 13 - SID Function Console Output



Item Code	Item Name	Item Brand	Price	Quantity	Category	Purchased Date
001	ZENBOOK	ASUS	250000.0	4	laptop	2025/03/20
002	air4pro	soundpeats	13000.0	6	erabuds	2024/04/30
022	iphone11	apple	180000.0	2	phone	2020/09/12

Figure 14 - SID Function TextFile Output

8.6SDD Function

Input

```
Type AID for adding item details.  
Type DID for deleting item details.  
Type UID for updating item details.  
Type VID for viewing the items table.  
Type SID for saving the item details to the text file.  
Type SDD for selecting four dealers randomly from a file.  
Type VRL for displaying all the details of the randomly selected dealers.  
Type LDI for displaying the items of the given dealer.  
Type ESC to exit the program.
```

```
Enter your choice: sdd
```

Figure 15 - SDD Function Input

Output

```
Enter your choice: sdd  
Selecting four dealers randomly from a file  
4 Dealers are Selected Randomly.
```

Figure 16 - SDD Function Output

8.7VRL Function

Input

```
Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.

Enter your choice: vrl
```

Figure 17 - VRL Function Input

Output

```
Enter your choice: vrl
Displaying all the details of the randomly selected dealers
--Dealer's Details--
| Name          | Contact No    | Location      |
| GigaTech      | 072-3344556   | Batticaloa   |
| TechWorld     | 071-2345678   | Colombo      |
| CyberLink     | 070-1122334   | Galle        |
| NetWorld      | 075-4455666   | Jaffna       |
```

Figure 18 - VRL Function Output

8.8LDI Function

Input

```
Type AID for adding item details.
Type DID for deleting item details.
Type UID for updating item details.
Type VID for viewing the items table.
Type SID for saving the item details to the text file.
Type SDD for selecting four dealers randomly from a file.
Type VRL for displaying all the details of the randomly selected dealers.
Type LDI for displaying the items of the given dealer.
Type ESC to exit the program.
```

```
Enter your choice: ldi
Display the items of the given dealer
Enter Dealer's Name : TechWorld
```

Figure 19 - LDI Function Input

Output

```
Enter Dealer's Name : TechWorld
--Dealer's Item Details--
| Name      | Brand      | Price(Rs.) | Quantity   |
| Laptop    | Dell       | 150000.0   | 5          |
| Mouse     | Logitech   | 3500.0     | 15         |
| Keyboard  | HP         | 4200.0     | 10         |
```

Figure 20 - LDI Function Output

8.9ESC Function

Input

```
Type AID for adding item details.  
Type DID for deleting item details.  
Type UID for updating item details.  
Type VID for viewing the items table.  
Type SID for saving the item details to the text file.  
Type SDD for selecting four dealers randomly from a file.  
Type VRL for displaying all the details of the randomly selected dealers.  
Type LDI for displaying the items of the given dealer.  
Type ESC to exit the program.  
  
Enter your choice: Esc
```

Figure 21 - ESC Function Input

Output

```
Enter your choice: Esc  
Have a nice day !!!  
  
Process finished with exit code 0  
|
```

Figure 22 - ESC Function Output